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3.1.1 INTRODUCTION

Concept Modelling technologies represent a class of AM processes primarily developed for rapid visualization, design verification, and functional prototyping. Unlike industrial-grade AM processes that focus on producing high-strength, fully dense parts, concept modellers emphasize speed, ease of use, surface finish, and visual fidelity of printed prototypes. These systems are extensively used in product design studios, research environments, and industries requiring quick turnaround times for concept evaluation and aesthetic assessment.

Concept modelling systems generally employ liquid photopolymers (or) thermoplastic materials, which are selectively deposited or solidified layer by layer to form three-dimensional models directly from digital CAD data. They offer high-resolution outputs, smooth surface finishes, and the capability to produce intricate geometries without complex post-processing requirements.

In AM, 3D printers are broadly categorized into Functional Modellers and Concept Modellers based on their purpose, precision, and application. Functional Modellers, often referred to as industrial-grade 3D printers, are designed to produce durable, high-performance components suitable for functional testing or direct end-use. In contrast, Concept Modellers are low-cost, quick-prototyping printers used primarily for visual or form-and-fit evaluations during the early stages of product design.

The following table shows the difference between Functional and Concept Modellers in terms of key characteristics and applications.

Difference between Functional and Concept Modellers in terms of key characteristics and applications

<i>Parameter</i>	Functional Modellers (Industrial-grade 3D Printers)	Concept Modellers (Low-cost, Quick Prototyping Printers)
<i>Purpose</i>	Used to produce fully functional prototypes and end-use parts	Used for visual and conceptual models to demonstrate design intent
<i>Accuracy</i>	Very high dimensional accuracy and surface finish	Moderate accuracy, suitable for design validation only
<i>Material Range</i>	Wide range includes engineering-grade polymers, composites, and metals	Limited to basic thermoplastics like PLA, ABS, or resin
<i>Build Volume</i>	Large build envelopes for industrial-scale components	Smaller build sizes, ideal for desktop applications
<i>Strength of Parts</i>	Parts exhibit high mechanical strength and thermal stability	Parts are weaker, mainly suitable for form and fit testing
<i>Cost</i>	High initial investment and operational cost	Low-cost and affordable for educational or design offices
<i>Speed</i>	Relatively slower due to precision and multi-step processes	Faster printing suitable for quick visualization
<i>Applications</i>	Aerospace, automotive, tooling, and functional testing	Product design visualization, education, and concept presentation

The present chapter will focus on the following concept modelling technologies:

- Thermal Jet Printer: Uses thermal inkjet printing technology to deposit droplets of thermoplastic or wax-based materials layer by layer. The material solidifies upon cooling, enabling the creation of detailed prototypes with smooth surfaces.
- Sanders ModelMaker 3D Printer: Employs precision inkjet heads to deposit build and support waxes alternately, achieving very fine resolution and dimensional accuracy. The support wax is later melted away to reveal the finished part, making it ideal for high-detail concept models and patterns for casting.
- Objet Quadra Systems: Utilize PolyJet technology, where multiple nozzles jet layers of liquid photopolymer that are instantly cured using ultraviolet (UV) light. This enables multi-material and multi-color printing with excellent surface finish and high dimensional accuracy.

3.1.2 THERMAL JET PRINTER

The Thermal Inkjet Printer, also known as a ThermoJet Printer, operates on the principle of Material Jetting (MJ) technology. In this additive manufacturing process, the system uses inkjet-style printheads to deposit microscopic droplets of molten thermoplastic or wax material onto a build platform, where each droplet rapidly cools and solidifies to form a layer. Simultaneously, a support material, typically a soluble wax is deposited to sustain overhangs and complex geometries during fabrication. This layer-by-layer deposition continues until the complete model is formed, producing parts with excellent surface finish and fine detail. However, the resulting components are primarily suitable for design verification or casting pattern applications, as the materials lack the mechanical strength required for functional use.

The Thermal Inkjet Printing approach was originally developed to create high-precision wax patterns for investment casting and Jewellery prototyping, where superior accuracy and surface smoothness are essential. The core technology, known as Thermal Phase Change Inkjet Printing, was pioneered by Solidscape, initially founded as Sanders Prototype, Inc. in 1993 by Royden C. Sanders in New Hampshire, USA. The company's mission was to design high-resolution wax printers for rapid prototyping and to produce master moulds for critical casting applications in the jewellery, dental, and precision engineering industries. Over time, the company's ownership evolved: in 2011, it was acquired by Stratasys Ltd., a global leader in polymer-based additive manufacturing; and in 2018, Stratasys sold Solidscape to the Prodways Group of France, enabling the brand to refine its specialization in industrial wax-based 3D printing solutions.

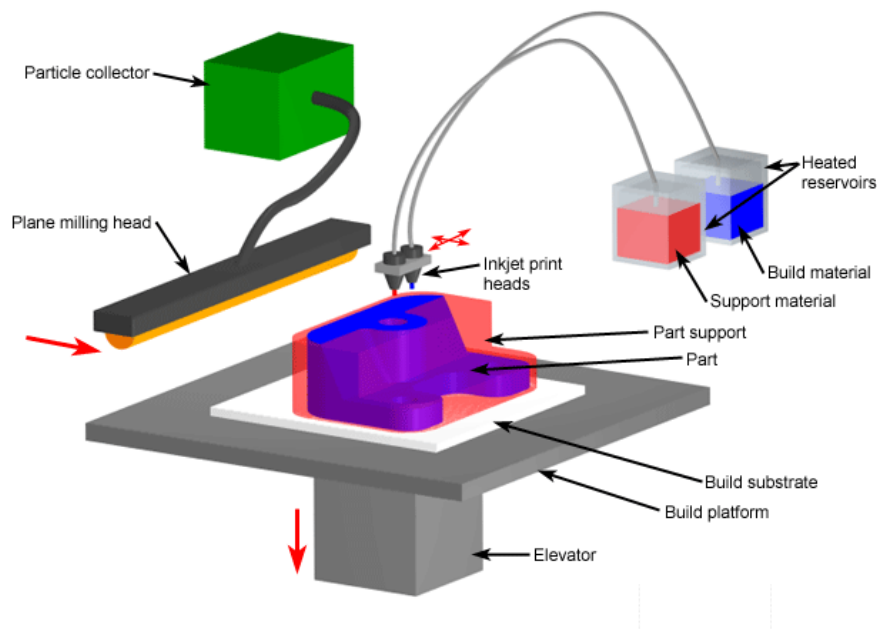
Thermal Inkjet Printing originates from 2D inkjet printer technology, where tiny droplets of liquid are ejected through nozzles to form images on paper. In 3D printing adaptation:

- The ink is replaced with molten thermoplastic or wax.
- The droplets are deposited layer by layer to form a 3D object.
- Upon deposition, the droplets instantly cool and solidify, forming a solid layer.

Because the material changes phase from liquid to solid during deposition, this process is often called “Thermal Inkjet Printing.”

HARDWARE

- Dual Printheads: One for build (model) wax and another for support wax. Both operate independently and eject molten material droplets in the X–Y plane.
- Heated Material Reservoirs (Cartridges): Maintain wax materials at precise temperatures to ensure correct viscosity for jetting. Separate reservoirs for model and support materials.
- Build Platform (XY Stage): A precision-controlled platform that moves horizontally to position droplets accurately during deposition.
- Elevator (Z-axis Mechanism): Lowers the platform after each completed layer to maintain consistent layer thickness.
- Plane Milling Head: A micro-milling cutter smooths each deposited layer, ensuring uniform thickness and high-quality surface finish.
- Particle Collector (Vacuum System): Removes wax debris generated by the milling process and stores it in a collection bin.



Schematic of Thermal Ink jet printer

WORKING

- The Thermal Inkjet Printing process, employed by SolidScape, utilizes both a model wax and a support wax, which are first melted in separate reservoirs for material preparation.
- The core of the build process is Droplet Deposition, where high-precision printheads traverse the build platform, accurately ejecting microscopic droplets of the molten wax materials at specific coordinates to define the part's current layer cross-section.
- These droplets immediately undergo Solidification as they rapidly cool upon contact with the previous layer or the platform, forming a stable structure.
- A milling head precisely trims the surface of the newly deposited layer to ensure exceptional flatness and a uniform height before the next layer is added. Concurrently, a Particle Removal system, typically a vacuum, collects and filters the fine wax dust generated by the milling process.

- Following these steps, the build platform executes an incremental Layer Advancement (Z-axis motion), preparing for the repetition of the entire process until the model is complete.
- Finally, post-processing involves easily melting away the distinct support wax, resulting in a pristine, high-resolution wax pattern that is immediately ready for use in investment casting.

ADVANTAGES

- Exceptional accuracy up to ± 0.025 mm.
- Smooth surface finish suitable for direct mould making.
- Ideal for intricate designs with fine details and internal cavities.
- Clean burnout of wax patterns leaves no ash or residue.

LIMITATIONS

- Slow build speed compared to photopolymer or powder-based systems.
- Limited to wax-like materials, restricting mechanical applications.
- Fragile printed parts not suitable for functional testing.
- High maintenance cost due to precise temperature control and milling systems.

APPLICATIONS

- Jewellery Manufacturing: Creation of highly detailed wax models for lost-wax casting. Enables intricate, customized designs with fine surface fidelity.
- Investment Casting: Production of clean-burning wax masters for precision casting of metals and alloys. Ideal for small, detailed components.
- Dental Applications: Fabrication of wax patterns for crowns, bridges, and dental restorations where dimensional precision is critical.
- Industrial Prototyping: Production of form-and-fit models for evaluating complex geometries before final tooling.

3.1.3 SANDERS MODELMAKER

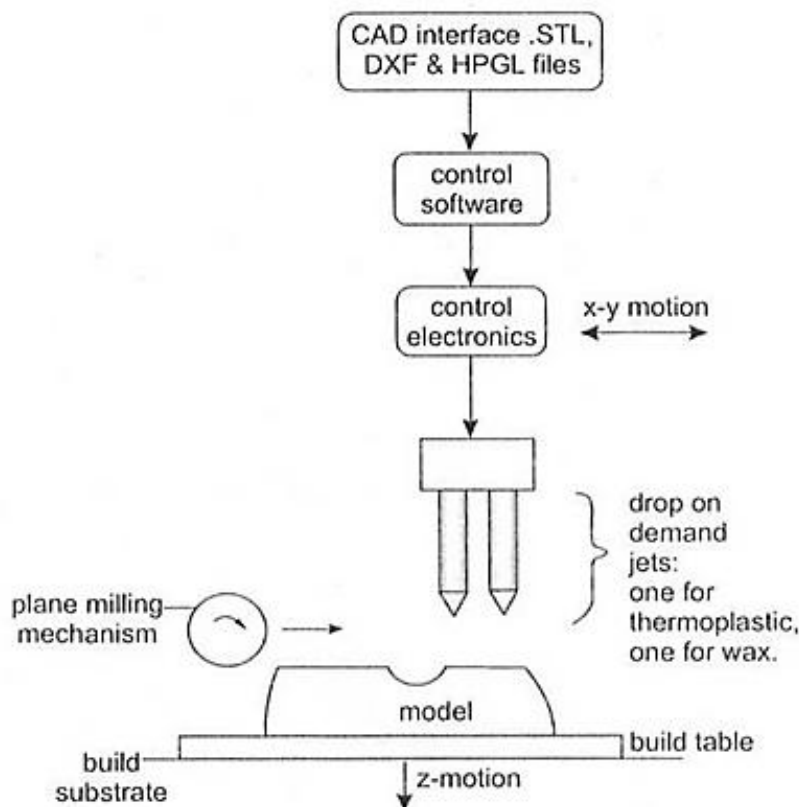
The Sanders Model Maker (MM) series represents one of the earliest and most precise implementations of inkjet-based additive manufacturing technology. Developed in the early 1990s by Sanders Prototype Inc., a subsidiary of Sanders Design in Wilton, New Hampshire, USA the Model Maker system was conceived to redefine accuracy and surface quality in rapid prototyping. Operating on the principle of layer-by-layer material jetting, the MM system deposits minute droplets of molten wax material to build complex three-dimensional geometries. Uniquely, the printer employs two separate inkjet heads one dedicated to the build material (model wax) and the other to a support material (sacrificial wax) both jetted concurrently to ensure dimensional stability and precise detailing, even in intricate or overhanging regions.

PRINCIPLE

The process starts by importing the 3D CAD model, which is sliced into thin layers by the system software. The machine heats the model and support wax materials to a precise

temperature and jets them alternately through fine drop-on-demand nozzles onto the build platform.

- The model wax forms the actual part geometry, while the support wax fills overhangs, cavities, or delicate features.
- After each layer is deposited, the material is solidified by cooling, ensuring a firm base for the next layer.
- This additive layering continues until the full 3D model is completed.
- Finally, the support wax is melted or dissolved away, leaving behind a smooth and highly detailed prototype ready for inspection or casting applications.



Schematic of Sanders Model Maker

ADVANTAGES

- Exceptional dimensional accuracy suitable for tooling and casting patterns.
- Produces parts with fine surface quality, reducing post-processing needs.
- Dual Material System – Simplifies the removal of support structures.
- Ideal for Investment Casting Patterns – Wax material makes it perfect for direct casting applications.

LIMITATIONS

- Layer-by-layer jetting and cooling make the process time-consuming.
- Restricted to wax materials, which are not suitable for functional testing.
- Wax models are delicate and require careful handling.
- Limits the size of parts that can be produced in a single build.

3.1.4 OBJET QUADRA SYSTEMS

The Objet Quadra System was one of the pioneering commercial 3D printers to implement what is now known as PolyJet technology, a process that combines inkjet printing principles with Ultraviolet (UV) photopolymer curing. Developed by Objet Geometries Ltd. In the early 2000s, the Quadra system was designed to produce high-resolution prototypes directly from CAD data, with exceptional surface finish and geometric accuracy. This innovation marked a major step in additive manufacturing, offering faster, cleaner, and more detailed prototyping than most competing technologies of its time.

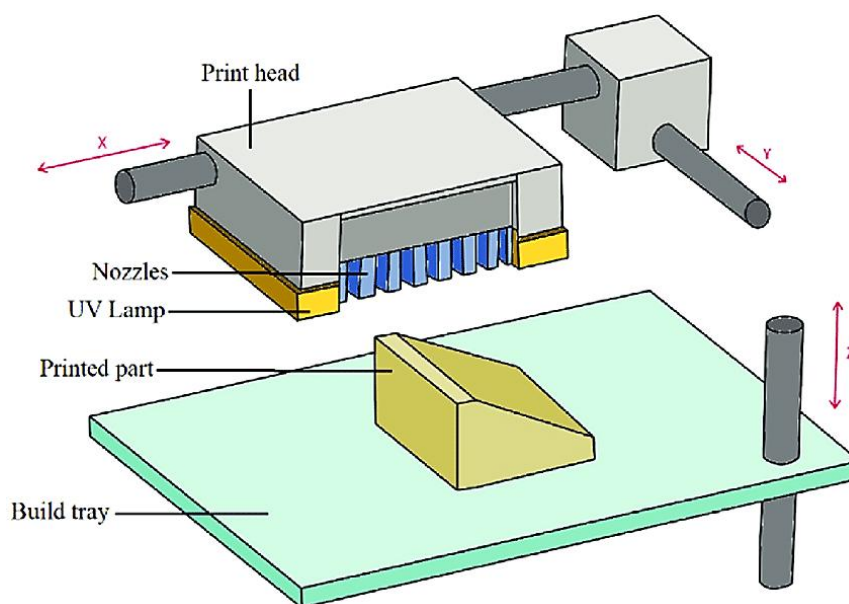
PRINCIPLE

The Quadra process operates on multi-nozzle inkjet printing, utilizing approximately 1,536 precision nozzles to jet ultra-thin layers of liquid photopolymer resin onto a build platform. Each layer is immediately cured by UV light, solidifying the resin before the next layer is deposited. This layer-by-layer curing process produces highly accurate, fully solid models without the need for post-curing or extensive post-processing.

Unlike traditional RP systems, the Quadra simultaneously jets two materials:

- Model material: A build resin forming the actual 3D geometry.
- Support material: A gel-like photopolymer that provides temporary support for overhangs and internal cavities.

Once printing is complete, the support material is manually or mechanically removed, leaving a smooth surface.



Schematic of Objet Quadra Systems

WORKING

The Objet Quadra process begins with importing the 3D CAD model, which is digitally sliced into thin layers to prepare it for printing. During fabrication, the system's printhead jets microscopic droplets of liquid photopolymer onto the build tray with exceptional precision. Simultaneously, integrated UV lamps instantly cure each deposited layer, solidifying the material as it is applied. This layer-by-layer fabrication continues until

the complete part is built. Once printing is complete, the support material is easily removed either manually or using water jets, revealing the final, high-precision model with a smooth surface finish and intricate details.

FEATURES

- High Resolution: Layer thickness as low as 16 microns (0.016 mm).
- Multi-Material Capability: Uses model and support materials simultaneously.
- Accurate and Smooth: Excellent dimensional accuracy and surface finish.
- Clean Process: Fully enclosed build chamber and automated operation.
- Multiple Jets: 1536 nozzles ensure fine detail and high build speed.

ADVANTAGES

- Exceptional accuracy and fine feature detail (± 0.1 mm typical).
- Smooth surfaces with minimal post-processing.
- Fast turnaround from digital model to physical prototype.
- Supports complex geometries, including internal cavities and thin walls.
- Efficient material handling through automated cartridge systems.

LIMITATIONS

- Restricted to photopolymer resins, which are less durable than engineering thermoplastics.
- Material cost and maintenance are relatively high.
- UV-cured materials may degrade or yellow under prolonged sunlight exposure.
- Limited mechanical strength and thermal resistance, restricting end-use applications.